



MUTHU MOHAMEDKABIR S M B

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PROFESSIONAL SUMMARY

Result-oriented B.E. Electrical and Electronics Engineering (EEE) final-year student (2026 Graduate) with a strong academic record. Eager to leverage technical expertise in **Embedded Systems** and **Software Engineering** to develop innovative hardware-software solutions. Proven capability in designing AI-driven integrations, full-stack web applications (MERN), and smart automation systems. Committed to contributing to technical R&D and engineering excellence in a challenging Graduate Engineering Trainee or Software Development role.

TECHNICAL SKILLS

Embedded & Engineering: Embedded Systems, Power Electronics, Electric Vehicle (EV) Fundamentals, HVAC Design, IoT Automation.

Software Engineering: Python, C#, C++, JavaScript (ES6+), C, SQL, REST APIs.

WebDevelopment: MERN Stack, Django, Flask, Flutter, HTML5, CSS3.

Hardware & Tools: Raspberry Pi, ESP32, Arduino, OpenCV, AutoCAD, MATLAB / Simulink, PCB Design, GitHub, VS Code.

ENGINEERING AND SOFTWARE PROJECTS

AI-Powered Sign Language Translation Platform

- Developed a real-time system using Python and Machine Learning to translate hand gestures into text/speech to assist the hearing impaired.
- Integrated MediaPipe and TensorFlow.js for high-accuracy gesture recognition and a MERN stack dashboard for user history.

IoT Smart Lock with Face Detection

- Designed and programmed a biometric security system using Raspberry Pi and OpenCV.
- Developed automated door control logic based on real-time facial recognition and integrated biometric authentication for enhanced security.

Smart Power Management System -

- Engineered an intelligent energy monitoring solution for Commercial Buildings using Non-Intrusive Load Monitoring (NILM).
- Integrated PZEM-004T sensors with ESP32 to capture real-time power data, resulting in optimized load pattern analysis.

2D CNC Printer Prototype

- Engineered a functional prototype for printing mini circuits, integrating electronics hardware with embedded software control.

PROFESSIONAL EXPERIENCE

MERN Stack Developer Intern | Codec Technologies FEB 2026

- Developed and maintained web applications using the MERN stack.
- Collaborated with the development team to implement RESTful APIs and responsive front-end components.

HVAC Design Intern | Bvisor

- Undergoing specialized training in HVAC system design, including heat load calculations and duct sizing.
- Gaining proficiency in interpreting architectural drawings and designing energy-efficient cooling solutions.

Intern Trainee (Electrical) | DCW Limited AUG 2025

- Gained hands-on experience in industrial electrical systems and maintenance workflows during a 3-week intensive program.
- Assisted senior engineers in routine inspections and power distribution analysis.

EDUCATION

B.E. in Electrical and Electronics Engineering PET Expected June 2026
Engineering College, Vallioor | Current CGPA: 7.61

Higher Secondary School Certificate (HSC) Graduated 2022
L.K. Higher Secondary School, Kayalpattinam

AWARDS AND CERTIFICATIONS

1st Prize: Circuit Debugging (National Level Symposium).
3rd Prize: Paper Presentation (National Level Symposium).
3rd Prize: Technical Quiz (National Level Symposium).
International Conference: Participated in International Conference held at PET Engineering College
Niral Thiruvizha 3.0: Team Leader of a state-wide Top 500 team for software innovation.
Certifications: AI, Energy Efficiency Building, PCB Design, Python & Django, React JS.

CO-CURRICULAR ACTIVITIES

Cultural Achievements: Won First Prize in multiple college-level singing competitions.
Continuous Learning: Proactive in self-improvement by exploring emerging technologies and learning through online resources.

LEADERSHIP AND LANGUAGES

Languages: English (Professional), Tamil (Native).
Leadership: Team Leader for multiple college technical projects and state-level hackathons.

I hereby declare that all the information provided above is true to the best of my knowledge

MUTHU MOHAMED KABIR
Signature.